



Lattice Boltzmann simulation of electrochemical systems[☆]

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Abstract

We propose a lattice Boltzmann model for studying electrochemical processes. A new lattice Boltzmann equation is introduced to simulate ion transports, including migration in an electric field, diffusion in a concentration gradient, and convection with fluid flow. The electrode reaction is incorporated via special boundary conditions for electric potential and ion flux. Benchmark studies on a variety of electrochemical problems, including the convective diffusion, the primary and secondary current distributions, and a general case, yield satisfactory results. © 2000 Elsevier Science B.V. All rights reserved.

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1. Introduction

Electrochemical processes are very important in both fundamental scientific research and industrial applications [1]. They are widely used in many material manufacturing and separation processes, such as electroplating, electrowinning, and electrorefining. Electrochemistry is the primary driving force in batteries and fuel cells. In addition, it plays a key role in many natural and industrial corrosion processes. In biological systems, most cells are living in an electrochemical environment. Studies of such important processes are of great significance.

A typical electrochemical system consists of many complex chemical and physical processes. Chemical reactions, especially those occurring at electrodes, are indispensable in the electrochemical processes. The migration of ions in the electric field between

electrodes is another process ubiquitous in electrochemistry. Yet, in most situations, electrochemical processes occur in a fluid environment, where diffusion and convection also play very important roles. In addition, solid surfaces in electrochemical systems, such as those on electrodes, usually undergo morphologic changes over time due to deposition or corrosion. Under certain circumstances, these changes lead to complex dendritic growth. Many of these processes are highly nonlinear and not fully understood. Theoretical analysis for this system are very difficult if not entirely impossible. Efficient numerical simulation tools are desired for both fundamental researches and industrial research and development.

Recent years have witnessed the fast development of a new numerical simulation tool, the lattice Boltzmann method (LBM), for transport phenomena (for a review, see [2]). The LBM simulates transport phenomena based on kinetic equations and statistical physics. Observations indicate that LBM could be an excellent candidate for studying electrochemical processes. First of all, it has been demonstrated by

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many authors that the lattice Boltzmann method is an accurate and efficient tool for simulating a variety of transport phenomena, including fluid flows [3–6], convection and diffusion of passive scalars [7,8], and chemical reactions [9–12]. More importantly, due to the kinetic nature, LBM has potentials to model chemical reaction kinetics at the microscopic or mesoscopic level [11,13]. Dendritic growth of deposits may be studied in a similar way to simulations of phase transition and multiphase flows [14–19]. Other advantages of the lattice Boltzmann method include its characteristics of parallel computing and the ease for programming.

Despite the above successes, there are several special issues that have not been well studied but are essential for us to establish a valid LBM model for electrochemical processes. First, we need to incorporate the electrical force to model the migration of ions. As shown in the following text, this can be done by introducing a new forcing term in the transport equation of ions. Secondly, we need to deal with the chemical reactions on electrode surfaces. These reactions are usually complex non-equilibrium processes. Although LBM has potential to model them at the mesoscopic or microscopic level, this study will neglect the details of the electrode reaction kinetics. Instead, the electrode kinetics is modeled by empirical laws such as Butler–Volmer law or its simplified versions, the linear polarization model and the Tafel model.

Besides the details of the electrode reaction kinetics, another issue that is skipped in this study is the movement of electrode surfaces due to the deposition of materials. This assumption is reasonable when the total deposition is small compared to the electrode size. Nevertheless, it precludes the studies of the dendritic growth. How to model the variation of electrode surface due to deposition and the consequent dendritic growth will be addressed in future studies.

The rest of the paper is organized as follows. Section 2 presents a framework for using lattice Boltzmann method to study electrochemistry. Section 3 discusses how to incorporate the electrode reaction into the model through boundary conditions. Section 4 presents benchmark studies to demonstrate the capability of the proposed LBM model for simulating a variety of electrochemical processes. Section 5 concludes the paper with discussions.

2. Lattice Boltzmann equations for electrochemistry

The lattice Boltzmann method simulates transport phenomena by tracking movements of molecule ensembles or the evolution of the distribution functions. With the assumption that the solute concentrations are sufficiently low not to influence the solvent flow, the solvent transport in two dimension can be described by the following lattice Boltzmann BGK equation [5, 6]:

$$f_\alpha(\mathbf{x} + \mathbf{e}_\alpha \delta_t, t + \delta_t) - f_\alpha(\mathbf{x}, t) = -\frac{f_\alpha(\mathbf{x}, t) - f_\alpha^{\text{eq}}(\rho, \mathbf{u})}{\tau_v + 0.5}, \quad (1)$$

where f_α is the density distribution function of the solvent, $\mathbf{x}$ is the space coordinates, δ_t is the time step, τ_v is the relaxation time, and f_α^{eq} is the equilibrium distribution of f_α :

$$f_\alpha^{\text{eq}}(\rho, \mathbf{u}) = w_\alpha \rho \left[1 + \frac{\mathbf{e}_\alpha \cdot \mathbf{u}}{RT} + \frac{(\mathbf{e}_\alpha \cdot \mathbf{u})^2}{2(RT)^2} - \frac{\mathbf{u}^2}{2RT} \right], \quad (2)$$

where R is the gas constant and T is the temperature. $\mathbf{e}_\alpha$'s are the discrete velocities and w_α 's are the associated weight coefficients. For the two-dimensional 9-speed LBM model, we have $RT = 1/3$ and,

$$\mathbf{e}_\alpha = \begin{cases} 0, & \alpha = 0, \\ (\cos[(\alpha - 1)\pi/2], \sin[(\alpha - 1)\pi/2]), & \alpha = 1, 2, 3, 4, \\ \sqrt{2}(\cos[(\alpha - 5)\pi/2 + \pi/4], \sin[(\alpha - 5)\pi/2 + \pi/4]), & \alpha = 5, 6, 7, 8. \end{cases} \quad (3)$$

The corresponding weight coefficients are $w_0 = 4/9$, $w_\alpha = 1/9$ for $\alpha = 1, 2, 3, 4$, and $w_\alpha = 1/36$ for $\alpha = 5, 6, 7, 8$. The density and velocity of the bulk fluid are calculated using:

$$\rho = \sum_{\alpha} f_\alpha, \quad (4)$$

$$\rho \mathbf{u} = \sum_{\alpha} f_\alpha \mathbf{e}_\alpha. \quad (5)$$

It has been rigorously approved that the above lattice Boltzmann equation recovers the Navier–Stokes equation for fluid flows near the incompressible limits. The kinematic viscosity is related to the relaxation time by $\nu = \tau_v RT$. The pressure can be calculated from fluid

density, $p = \rho RT$. Numerical studies have verified the theoretical prediction.

The eventual goal of electrochemistry is to study the transport of ions. In general, ions can be transported by the convection of fluids, by the diffusion in a concentration gradient, and by the migration in an electric field. Using LBM to simulate convection and diffusion of solutes has been studied before (e.g., [7, 8]). In this paper, we proposed to incorporate further the ion migration by adding a new electric force term which is similar to the intermolecular force in multiphase flow [17]. The proposed evolution equation for ion transport has the following form:

$$\begin{aligned} g_{\alpha}^i(\mathbf{x} + \mathbf{e}_a \delta_t, t + \delta_t) - g_{\alpha}^i(\mathbf{x}, t) \\ = -\frac{g_{\alpha}^i(\mathbf{x}, t) - g_{\alpha}^{i,\text{eq}}(c_i, \mathbf{u})}{\tau_i + 0.5} \\ + \frac{z_i F (\mathbf{e}_a - \mathbf{u}) \cdot \nabla \phi}{RT} g_{\alpha}^{i,\text{eq}}, \end{aligned} \quad (6)$$

where g_{α}^i is the distribution function of species i and τ_i is a relaxation time. $g_{\alpha}^{i,\text{eq}}$ is the corresponding equilibrium distribution:

$$g_{\alpha}^{i,\text{eq}}(c_i, \mathbf{u}) = w_{\alpha} c_i \left[1 + \frac{\mathbf{e}_a \cdot \mathbf{u}}{RT} + \frac{(\mathbf{e}_a \cdot \mathbf{u})^2}{2(RT)^2} - \frac{\mathbf{u}^2}{2RT} \right]. \quad (7)$$

The concentration of species i is calculated using:

$$c_i = \sum_{\alpha} g_{\alpha}^i. \quad (8)$$

Using the Chapman–Enskog expansion technique, we can prove that the above lattice Boltzmann equation for ion transport recovers the following macroscopic equation:

$$\begin{aligned} \frac{\partial c_i}{\partial t} + (\mathbf{u} \cdot \nabla) c_i \\ = \nabla \cdot (D_i \nabla c_i) + \nabla \cdot \left[\frac{z_i D_i F c_i}{RT} \nabla \phi \right], \end{aligned} \quad (9)$$

where the diffusion coefficient, D_i , is related to the corresponding relaxation time via $D_i = \tau_i RT$.

The second term in the right hand side of Eq. (6) warrants some special discussions. It represents the electric force which drives the migration of ions, where z_i is the charge number of the ion and F is the Faraday constant. The electric potential, ϕ , must

be determined by additional physical laws. In electrochemistry, it is usually assumed that the solution is electrically neutral, i.e. the charges on the cations are balanced by those on the anions. This implies that the ionic concentrations are related by:

$$\sum_i z_i c_i = 0. \quad (10)$$

This neutrality law leads to the following governing equation for the electric potential:

$$\nabla \cdot (k \nabla \phi) + F \nabla \cdot \sum_i (z_i D_i \nabla c_i) = 0, \quad (11)$$

where $k = F^2 \sum_i z_i^2 D_i c_i / RT$ is the electric conductivity. Eq. (11) is an elliptic partial differential equation and can be solved using any Poisson solvers. In the framework of LBM, the solution of Eq. (11) can be obtained as the steady state solution of the following lattice Boltzmann equation:

$$\begin{aligned} h_{\alpha}(\mathbf{x} + \mathbf{e}_a \delta_t, t + \delta_t) - h_{\alpha}(\mathbf{x}, t) \\ = -\frac{h_{\alpha}^i(\mathbf{x}, t) - h_{\alpha}^{\text{eq}}(\phi)}{\tau_{\phi} + 0.5} \\ + w_{\alpha} F \nabla \cdot \sum_i (z_i D_i \nabla c_i), \end{aligned} \quad (12)$$

where $h_{\alpha}^{\text{eq}}(\phi) = w_{\alpha} \phi$ and $\tau_{\phi} = k / RT$.

3. Modeling electrode reaction in boundary condition

In the last section, we have formulated a framework for using the lattice Boltzmann method to simulate electrochemical transport process. So far, however, we have not included any chemical reactions. The chemical reaction in electrochemistry is unique in that it occurs in a thin double charged layer between electrode and electrolytic solution. The thickness of the double layer is between 10 to 100 Å and often negligible in practical applications. Because of this, it is reasonable to assume that the chemical reaction in electrochemistry occurs only at electrode surfaces. Consequently, the reaction should be dealt with through boundary conditions for ion transport.

The electrode reaction is a complicated process. In general, it can be expressed as:



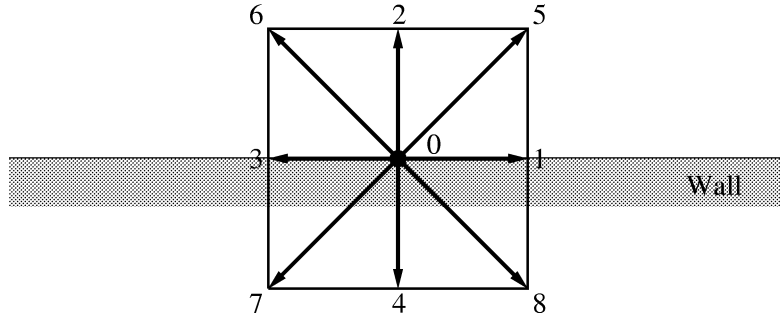


Fig. 1. A wall node at the bottom surface.

where s_i is the stoichiometric coefficient of species i in the electrode reaction, M_i represents the chemical composition of species i , e^- represents an electron, and n is the number of free electrons generated or annihilated in the electrode reaction. One of the important quantities in the electrode reaction is the equilibrium potential when the reaction reaches an equilibrium state. It can be expressed as:

$$\mu_e^0 = \sum_i \frac{s_i \mu_i}{n}, \quad (14)$$

where μ_i is the electrochemical potential of the charged species i .

In most electrochemical systems, the electrode reaction is a transient process. The reaction rate, characterized by the current density, depends on a variety of factors such as the nature and property of electrode surfaces, the composition of the electrolytic solution, and the electrode potential. Experimental results show that many electrode reactions can be described by the following Butler–Volmer law:

$$i_n = i_0 \left[\exp\left(\frac{\alpha_a F}{RT} \eta_s\right) - \exp\left(-\frac{\alpha_c F}{RT} \eta_s\right) \right], \quad (15)$$

where i_n is the current density; i_0 is the exchange current density which depends on the composition of the electrolytic solution adjacent to the electrode; α_a and α_c are transfer coefficients which are usually determined by experimental measurements; η_s is the surface overpotential which is the departure of the electrode potential from the equilibrium potential, $\eta_s = \phi - \mu_e^0$. The first term in the Butler–Volmer law represents the rate of the anodic process, and the second term, that of the cathodic process.

The Butler–Volmer law can be simplified in two limit conditions. When the surface overpotential is small, it can be approximated by the following linear polarization model:

$$i_n = i_0 \frac{(\alpha_a + \alpha_c) F}{RT} \eta_s. \quad (16)$$

When the surface overpotential is large, it can be simplified to the following Tafel model (at the anode, for an example):

$$i_n = i_0 \exp\left(\frac{\alpha_a F}{RT} \eta_s\right). \quad (17)$$

The above electrode reaction laws provide boundary conditions for both electric potential and ion transport. The boundary condition for electric potential can be written as:

$$k \frac{\partial \phi}{\partial n} = i_n, \quad (18)$$

where k is the conductivity of electrolytic solution near the electrode. Since in general the current density is a nonlinear function of the surface potential, it is cumbersome to solve the potential field especially with analytical methods. Previous approaches rely, to a large extent, on the successive relaxation method [1].

This dilemma is relatively easy to handle in the framework of the lattice Boltzmann method. This is because the boundary condition in LBM is applied to the incoming distribution functions instead of the potential and current density themselves. Taking a boundary node at a horizontal bottom boundary as an example (Fig. 1). All we need to know are the distribution functions h_2 , h_5 , and h_6 . According to

kinetic theory [20,21], the distribution function can be approximated as:

$$h(\mathbf{x}, \xi, t) = h^{\text{eq}}[1 - \tau_\phi(\xi \cdot \nabla) \ln \phi], \quad (19)$$

or in the discrete form:

$$h_\alpha = h_\alpha^{\text{eq}}[1 - (\tau_\phi + 0.5)(\mathbf{e}_\alpha \cdot \nabla) \ln \phi]. \quad (20)$$

In Eq. (20), $\tau_\phi + 0.5$ replaces τ_ϕ as the coefficient of the non-equilibrium part of the distribution function. The additional 0.5 comes from the discretization procedure. Although Eq. (20) gives the distribution function as a function of the potential ϕ and the ion flux $\nabla\phi$, it cannot be used directly because both potential and flux are not explicitly known.

In this study, we propose to use an alternative procedure. From Eq. (20), it is straightforward to find:

$$h_2 + h_5 + h_6 + h_4 + h_7 + h_8 = \frac{\phi}{3},$$

$$h_2 + h_5 + h_6 - h_4 - h_7 - h_8 = -\left(1 + \frac{1}{2\tau_\phi}\right)i_n,$$

where $i_n = k\partial\phi/\partial n$ is the current density. Together, these two equations lead to:

$$\phi + 3\left(1 + \frac{1}{2\tau_\phi}\right)i_n = 6(h_4 + h_7 + h_8). \quad (21)$$

With h_4 , h_7 , and h_8 known, ϕ can be calculated from Eq. (21) for any given electrode reaction. Eq. (21) can be readily solved using Newton–Raph interaction scheme. Once ϕ is known, the incoming distribution function can be readily calculated using:

$$h_2 = (h_2^{\text{eq}} + h_4^{\text{eq}}) - h_4, \quad (22)$$

$$h_5 = (h_5^{\text{eq}} + h_7^{\text{eq}}) - h_7, \quad (23)$$

$$h_6 = (h_6^{\text{eq}} + h_8^{\text{eq}}) - h_8. \quad (24)$$

These equalities are based on the fact that the sum of two distribution functions along two opposite directions equals to the sum of their equilibrium distribution functions. They can be derived directly from Eq. (20).

The boundary condition of ion transport is relatively simple. The no-flux boundary condition applies to non-reacting species. For reacting species, ion flux across the boundary can be calculated from the current density. Similarly, the ion concentration at boundary can be calculated as:

$$c_i = 6(g_4^i + g_7^i + g_8^i) - 3\left(1 + \frac{1}{2\tau_i}\right)\frac{s_i i_n}{z_i F}, \quad (25)$$

and the incoming ion flux can be calculated using:

$$g_2^i = (g_2^{i,\text{eq}} + g_4^{i,\text{eq}}) - g_4^i, \quad (26)$$

$$g_5^i = (g_5^{i,\text{eq}} + g_7^{i,\text{eq}}) - g_7^i, \quad (27)$$

$$g_6^i = (g_6^{i,\text{eq}} + g_8^{i,\text{eq}}) - g_8^i. \quad (28)$$

4. Benchmark studies

To test the proposed LBM model, we have carried benchmark studies for a number of electrochemical problems, including the convective diffusion problem, the primary and secondary current distributions, and a general case. These test cases cover a wide range of applications varying from diffusion-dominant problems to migration-dominant ones. For each case, the computational grid was refined until a converged result was obtained.

4.1. Case one – convective diffusion of ions

For the reaction of one minor ionic species in a solution containing excess supporting electrolyte, the ion migration is negligible. The ion transport is primarily controlled by diffusion and convection. A typical example is a flowing channel with ion flowing in at a given concentration while the concentration is fixed at another constant value at the bottom surface (Fig. 2). Frequently, the ion diffusion coefficient is small and

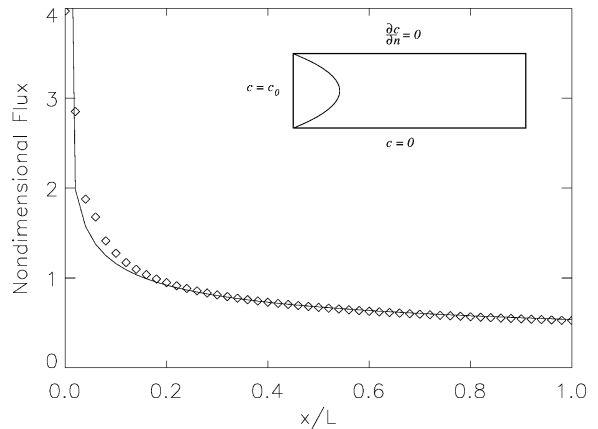


Fig. 2. Comparison of the calculated ion flux at the surface (symbols) with the L  v  que solution (solid line). The legend shows the experimental setup.

the concentration variation is confined in a thin boundary layer near the surface. For laminar flow in which the velocity assumes a parabolic profile, there exists an analytic solution – the L  v  que solution – for this classical convective diffusion problem [22]. Fig. 2 shows the ion flux across the bottom surface as a function of the distance from the inlet of the channel. The solid line is the L  v  que solution while the symbols are results from our LBM simulation. We have found good agreement between the two results except around the inlet corner, where the theoretical concentration gradient possess a singularity.

4.2. Case two – migration of ions

In contrast to the process in the above convective diffusion problem, the ion diffusion coefficient is large and the fluid flow is fast in some other electrochemical systems. Consequently, the variation in ion concentration is negligible and the ion migration has the dominant effect. The proposed theory is applicable to this type of problems as well.

One typical ion migration problem occurs when the surface potential is low and the polarization of the electrode is negligible. In this scenario, the electrode can be regarded as an equipotential surface. The corresponding current distribution is the so-called “primary current distribution”. To check the accuracy of our LBM model, we choose to simulate the potential field in an insulating channel with two plane electrodes placed opposite each other. Fig. 3 shows the geometric setup of the reacting chamber. The simulations were carried out on a quarter of the reaction chamber and symmetry was applied at the two mid-planes. The current distribution on the electrode surface is shown in

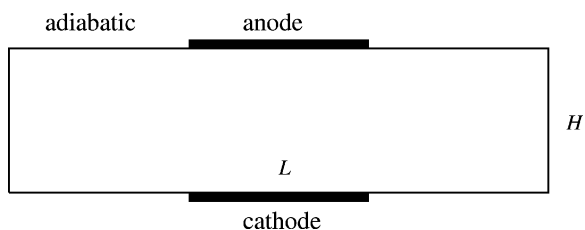


Fig. 3. Geometric setup for reaction chamber.

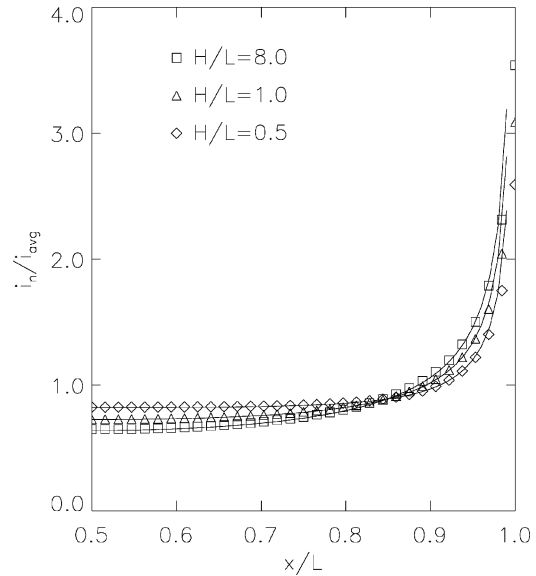


Fig. 4. Comparison of primary current distribution on channel electrodes, LBM (symbols) vs theory (solid lines). The horizontal axis is the non-dimensional distance to the left edge of the electrode.

Fig. 4. The LBM simulation results agree well with the theoretical predictions [1]:

$$\frac{i_n}{i_{\text{avg}}} = \frac{\epsilon \cosh \epsilon / K (\tanh^2 \epsilon)}{\sqrt{\sinh^2 \epsilon - \sinh^2 (2x/L - 1) \epsilon}}, \quad (29)$$

where $\epsilon = \pi L/2H$ with H being the height of the channel and L the width of the electrode [1]. x is measured from the edge of the electrode and $K(m)$ is the complete elliptic integral of the first kind. Notice that the theoretical prediction of the current tends to infinity at the electrode edges. That explains the discrepancy between the numerical and theoretical results at the electrode edges.

In many scenarios, electrode polarization is not negligible and the electrode reaction has to be taken into account. This yields the so-called “secondary current distribution”. In some cases, the surface overpotential is small and the electrode kinetics can be described by the linear polarization model Eq. (16). Fig. 5 shows the LBM-simulated secondary current distributions along the electrode in the same reaction chamber. They agree well with the simulation results using the finite element method [23].

The polarization of the electrode tends to eliminate the infinite current density at the edges of electrodes

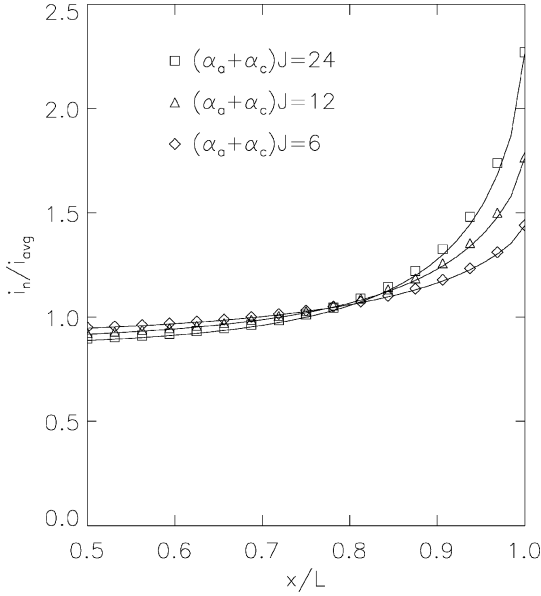


Fig. 5. Comparison of secondary current distribution for linear polarization, LBM (symbols) vs finite element analysis (solid lines). The channel aspect ratio is $H/L = 0.5$. The horizontal axis is the non-dimensional distance to the left edge of the electrode.

and to make the secondary current distribution more uniform. The uniformity of the current distribution can be characterized by the ratio of the maximum current at the edge to the minimum current at the center. Fig. 6 shows the ratio of maximum to minimum current as a function of the non-dimensional exchange current density:

$$(\alpha_a + \alpha_c)J = \frac{i_0(\alpha_a + \alpha_c)FL}{RTk}. \quad (30)$$

Our LBM simulation agrees with the finite element analysis in a wide parameter range.

In other cases, the surface overpotential is large and the electrode kinetics can be described by the non-linear Tafel model Eq. (17). This nonlinear boundary problem can be solved using our LBM model. Fig. 7 shows the ratio of maximum to minimum current as a function of the non-dimensional average current:

$$\delta = \frac{|i_{\text{avg}}|FL}{RTk}. \quad (31)$$

The simulated results agree well with the previous studies by Parrish and Newman [24].

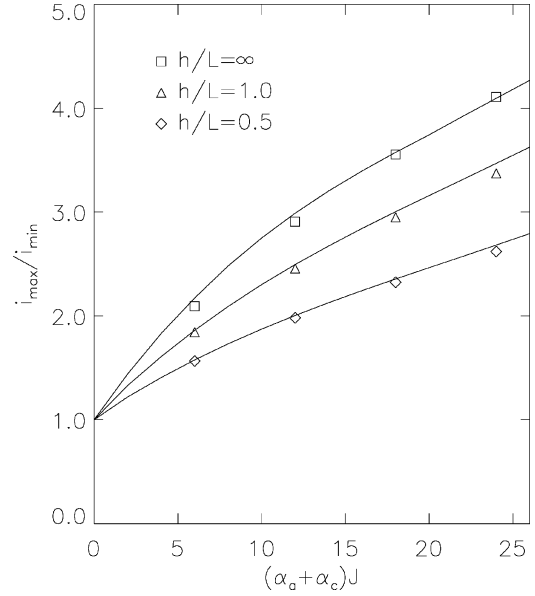


Fig. 6. Ratio of maximum to minimum current for linear polarization as a function of the non-dimensional exchange current. The symbols are results from LBM analysis and the solid lines are results from finite element analysis. The horizontal axis is the non-dimensional exchange current.

4.3. Case three – general condition

To appreciate the complexity of the electrochemical process, we choose in our final benchmark study to simulate a general electrochemical processes with migration, diffusion, convection, and electrode reaction. The geometric setup is the same as in Fig. 3. The electrolytic solution flows in the channel with the ion concentration fixed at c_0 . The electrode reaction at the cathode follows the linear polarization law, Eq. (16). We assume the polarization at anode is negligible and the anode is an equipotential surface. The ion flux across the anode is also negligible.

One of the characteristic parameters in this electrochemical process is the non-dimensional potential difference between two electrodes:

$$N = \frac{(\alpha_a + \alpha_c)F(\phi_a - \mu_e^c)}{RT}, \quad (32)$$

where μ_e^c is the equilibrium potential at the cathode. When the potential difference is low, the ion migration is dominant and the current distribution is similar to the secondary current distribution. The migration of ions in this limiting case has minor effects and

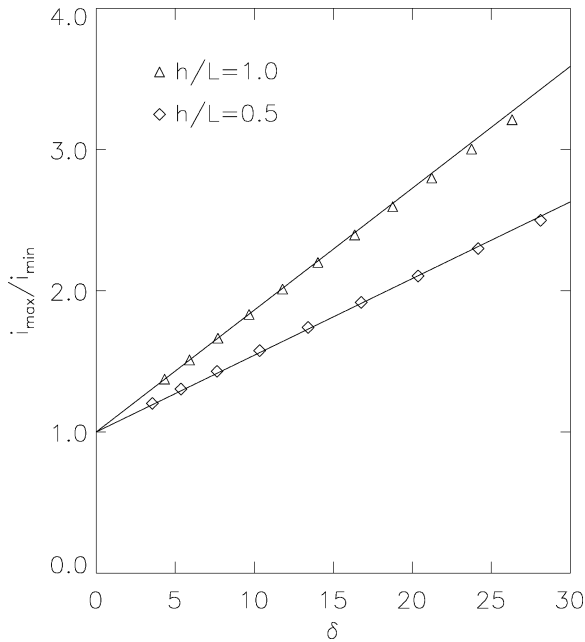


Fig. 7. Ratio of maximum to minimum current for Tafel polarization model as a function of the non-dimensional average current. The symbols are results from LBM analysis and the solid lines are results by Parrish and Newman [24]. The horizontal axis is the non-dimensional average current.

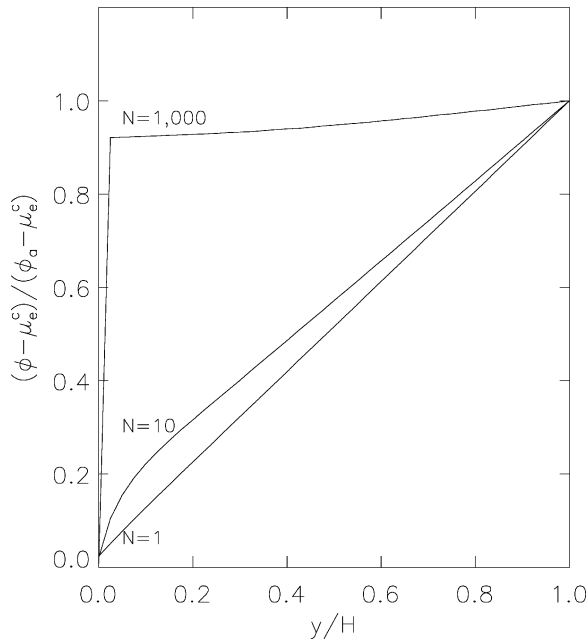


Fig. 8. Potential profiles cross the channel at the channel center.

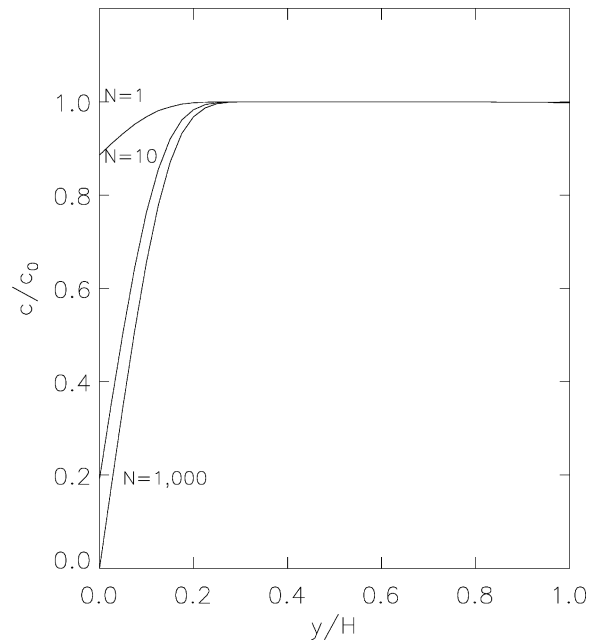


Fig. 9. Concentration profiles cross the channel at the channel center.

the concentration variation in the channel is negligible. On the other hand, when the potential difference is high, the ion migration causes the depletion of ions near the cathode. A concentration boundary layer exists very much like that in the convective diffusion problem. The potential variation is small in the bulk solution. For moderate potential difference, both potential and concentration variations are important. Our LBM simulations successfully capture these phenomena, as shown in Fig. 8 for the potential profiles and in Fig. 9 for the concentration profiles across the channel at the channel center.

5. Conclusions

In conclusion, we have developed a lattice Boltzmann model for studying electrochemical system. The fluid flow is simulated using the standard LBM-BGK models. A new LBM equation is introduced to simulate ion transports, including migration, diffusion, and convection. The electric potential is also solved using a separated LBM equation. The electrode reaction is incorporated via special boundary conditions for electric potential and ion flux. Although this study only

focuses on two-dimensional problems, the formulation can be easily extended to three-dimension.

Benchmark studies have been carried out for a variety of electrochemical problems, ranging from the diffusion-dominant convective diffusion to the migration-dominant primary and secondary current distributions. Comparisons with theory and other previous numerical studies yield satisfactory results. The success of these benchmark studies provides us confidence in using this LBM model to study other electrochemical problems.

One issue that remains to be explored is how to solve the potential field which satisfies an elliptic partial differential equation. Theoretically, any Poisson solver can be used for this purpose. This study uses an LBM Poisson solver which is easy to apply in complicated geometries such as those in dendritic growth, but relatively slow compared to other Poisson solvers such as the alternative direction iteration (ADI) and the conjugate gradient method (CGM). It is desirable to design a faster Poisson solver that can handle complicated geometry. In the future, we would also like to extend the proposed LBM model to study dendritic growth.

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